

Haibiao Zhang

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[Google Scholar](#) | [GitHub](#)

BIOGRAPHY

Ph.D. candidate in Computer Technology at the University of Science and Technology of China (USTC), working on artificial intelligence for large engineering systems, with a focus on data-driven modeling and fault prediction, Transformer-based time-series analysis, and AI proof-of-concept applications in nuclear-fusion engineering systems.

EDUCATION

University of Science and Technology of China (USTC) [\[link\]](#)

Ph.D. Candidate in Computer Technology

Sep 2021 – Present

Hefei, China

- Research Focus: AI-driven modeling and fault prediction for gyrotron-based heating and current drive systems

Southwest Forestry University (SWFU) [\[link\]](#)

B.E. in Communication Engineering

Sep 2017 – Jul 2021

Kunming, China

- GPA: 3.79/4.00 (Rank 1/36)

RESEARCH OUTPUTS

C=CONFERENCE, J=JOURNAL, P=PATENT, SW=SOFTWARE COPYRIGHT, S=IN SUBMISSION

- [J.1] Haibiao Zhang, Zege Wu, Huaichuan Hu, and Xiaojie Wang (2024). **Data-driven fault diagnosis method for abnormal RF oscillation of gyrotrons**. *AIP Advances*, 14(4), 045320. DOI: 10.1063/5.0195400
- [J.2] Chen Wu, Haibiao Zhang, Haoming Chang, Jingping She, and Xiaojie Wang (2025). **Robust learning-based fault prediction method for gyrotron system**. *AIP Advances*, 15(3), 035150. DOI: 10.1063/5.0257521
- [J.3] Xiaoguang Zhang, Nengtao Zhou, Haibiao Zhang, Dingzhen Li, Yu Hua, Chao Xu, Chao Zhang, Ming Gao, Xiaojie Wang, and Yunying Tang (2025). **Simulation analysis of the motion accuracy of the ECRH launcher steering mechanism based on ADAMS**. *Fusion Engineering and Design*, 211, 114774. DOI: 10.1016/j.fusengdes.2024.114774
- [J.4] Yuanyuan Kuang, Hanlin Wang, Ming Gao, Chao Xu, Haibiao Zhang, Yaopeng Dai, Yunying Tang, and Xiaojie Wang (2025). **Investigation of electron cyclotron wave absorption and current drive in CFEDR conventional H-mode scenario**. *Plasma Science and Technology*, 27(10), 104003. DOI: 10.1088/2058-6272/ade423
- [S.1] Haibiao Zhang, et al. (2026). **TWCS: A Tolerance-Aware Workflow for Conformal Screening of Gyrotron Electron-Gun Designs in Fusion Heating Systems**. Manuscript submitted to *AAAI 2027* (under review).
- [P.1] Invention Patent (Granted): Distributed Control and Dynamic Scheduling Method and System for ECRH Systems (CN 120029768 B)
- [P.2] Invention Patent (Granted): Fast Voltage Regulation Method and System for Gyrotrons Based on Long-Pulse Power Forbidden-Zone Prediction (CN 121619724 B)
- [SW.1] Software Copyright: Gyrotron Operating-State Probability Prediction System V1.0 (2025SR1824848)
- [SW.2] Software Copyright: Comprehensive Signal Detection and Analysis Software for Communication Equipment V1.0 (2020SR0059618)

HONORS AND AWARDS

- First-Class Academic Scholarship, University of Science and Technology of China (USTC) 2021 – 2025
- Yunnan Provincial Outstanding Graduate Jun 2021
- National Scholarship Dec 2019
- National Second-Class Award, National Undergraduate Mathematical Modeling Competition (NUMMC) Nov 2019

PROJECT EXPERIENCE

Data-driven Optimization of Electron Gun Parameters and Operating Boundaries

Institute of Plasma Physics, Chinese Academy of Sciences | Advisor: Prof. Xiaojie Wang

Sep 2024 – Dec 2025

Hefei, China

- Developed data-driven AI models to identify safe operating boundaries of electron-gun parameters for fusion heating system design
- Combined machine learning with sensitivity analysis and a targeted sampling strategy, reducing 3D charged-particle simulation cost by approximately 50%

- Translated model predictions into quantified engineering constraints to support design optimization and operational decision-making
- Proposed the Trajectory-Witness Conformal Surrogate (TWCS) framework for tolerance-aware screening of electron-gun designs; manuscript under review at AAAI 2027
- Collaborated closely with physics and mechanical engineering teams to ensure AI predictions remain consistent with physical laws and engineering constraints

Fault Prediction of Gyrotron Operating Parameters

Sep 2022 – Jun 2024

Institute of Plasma Physics, Chinese Academy of Sciences | Advisor: Prof. Xiaojie Wang

Hefei, China

- Curated a dataset of 22,000+ operational configurations from gyrotron systems as a foundation for fault prediction
- Designed Transformer-based and ensemble learning models for time-series prediction and fault-state estimation, enabling probabilistic assessment of equipment health
- Benchmarked KNN, Random Forest, and other models to improve the accuracy and robustness of fault identification under complex parameter interactions
- Related results published in *AIP Advances*

EXTRA-CURRICULAR ACTIVITIES

Internship at CCTEG Changzhou Research Institute

Aug 2025

- Completed the three-level safety education training and assessment, systematically learning production protection and emergency response protocols
- Engaged in technical discussions with the institute’s R&D team on AI large-model empowerment of integrated mine monitoring and control systems, focusing on real-time optimization via algorithm lightweighting and edge computing deployment
- Investigated efficient vector-encoding methods for multi-source heterogeneous monitoring data to support accurate perception of dynamic mine operating environments

ECRH System Experimental Operation (CRAFT Fusion Reactor)

Sep 2022 – Present

- Participated in installation, commissioning, and operation of ECRH systems
- Collected experimental data and investigated fault phenomena during system operation
- Prepared technical test reports to support troubleshooting and system reliability analysis

AI & Nuclear Fusion Science Popularization Volunteer

Jun 2023 – Sep 2023

- Organized science popularization events targeting primary and secondary school students, reaching over 1,000 participants in total
- Accumulated 300+ volunteer hours; awarded “Outstanding Volunteer” by Anhui Women and Children’s Activity Center and “One-Star Volunteer” by the University of Science and Technology of China (USTC)

SKILLS

- **Programming & AI:** Python (NumPy, SciPy, scikit-learn), PyTorch, TensorFlow, C
- **Engineering Tools:** MATLAB, CST Particle Studio, Git, Docker, Data Visualization
- **Instrumentation:** Spectrum Analyzer, Vector Network Analyzer, Oscilloscope, Thermal Imager
- **Languages:** English (CET-6, LSCAT Translation Level 1)